

Received-Power-Aware Carrier Phase Estimator Selection for Turbulent Satellite–Ground FSO Links

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Abstract—Coherent free-space optical links require carrier recovery that remains effective as received power changes with atmospheric turbulence. Data-aided (DA) and non-data-aided (NDA) phase estimators offer complementary operating regions: pilots support phase estimation when the effective signal-to-noise ratio (SNR) is low, while the NDA phase statistic preserves every transmitted position for data. This paper develops a two-stage selector that first detects strongly varying received-power windows through a coefficient-of-variation gate and then combines a received-power-based irradiance proxy with a fixed 13 dB effective-SNR decision level. The selector chooses between multi-pilot phase-time least squares and an eighth-power NDA estimator. Simulations of 16-ary amplitude-phase shift keying (16APSK) over additive-noise and five Gamma–Gamma turbulence conditions characterize the complementary estimator behavior using per-window ambiguity-resolved NDA evaluation. In the predefined high-SNR work region, fixed NDA provides up to a 3.1 dB total-energy-equivalent advantage over fixed DA; this value comprises a 1.9 dB data-symbol-SNR gap and the approximately 1.25 dB pilot-energy term of the DA branch. The results demonstrate that a fixed, interpretable decision rule based on received-power statistics can support estimator selection across changing atmospheric conditions.

I. INTRODUCTION

The continued deployment of low-Earth-orbit (LEO) satellite constellations is increasing the capacity required from feeder links between satellites and optical ground stations [1]. Coherent free-space optical (FSO) communication is attractive for this role because coherent reception offers high sensitivity, supports spectrally efficient higher-order modulation, and enables digital compensation of carrier impairments [2]–[4]. The atmospheric path nevertheless introduces absorption, scattering, and refractive-index fluctuations. Turbulence produces scintillation in the received irradiance and perturbs the optical phase, while satellite motion and finite laser linewidth add time-varying carrier-frequency and phase errors [3], [5], [6]. These effects make carrier phase recovery (CPR) a central receiver function [7]: the residual phase must be estimated accurately enough to preserve the decision regions of a high-order constellation as the received signal quality changes over time. Recent coherent-FSO studies consequently combine atmospheric mitigation with digital carrier synchronization or robust data-aided processing [3], [8], [9].

Data-aided (DA) and non-data-aided (NDA) carrier estimators provide complementary ways to obtain that phase reference. A DA receiver inserts known pilot symbols, allowing the receiver to form direct phase observations and fit the carrier evolution from regularly spaced references. Data-aided processing has demonstrated stable coherent reception at very low signal-to-noise ratios (SNRs) in free-

space experiments [8]. An NDA phase statistic instead removes the modulation phase from the received data. The classical Viterbi–Viterbi construction raises phase-shift-keyed samples to the rotational-symmetry order before averaging their phases [10]; the same principle extends to amplitude-phase-shift-keyed constellations by using the number of phase points on each ring [11], [12], building on joint maximum-likelihood phase estimation theory developed for sinusoidal carriers under Wiener phase noise [13], [14]. NDA therefore retains every transmitted position for payload and can aggregate the full observation window, whereas DA obtains explicit references at a prescribed pilot density. The reported NDA bit-error evaluation resolves the resulting rotational ambiguity per processing window using the transmitted bit labels, while the selector itself uses only pre-recovery received-power statistics. Under block-varying Gamma–Gamma irradiance, the effective SNR seen by successive processing windows changes with the atmospheric gain. The two reference strategies can consequently favor different operating regions, creating a direct motivation for receiver-side selection across these regions, an approach also explored in pilot-aided adaptive receiver designs [15].

This paper develops a blockwise DA/NDA selector for a coherent satellite–ground FSO link using (8, 8) 16-ary amplitude-phase shift keying (16APSK). The selector applies one fixed two-stage decision procedure to every 256-sample digital-signal-processing window. First, a coefficient-of-variation boundary obtained from offline AWGN received-power characterization screens the raw received-power samples. For the remaining windows, a received-power-based channel proxy is converted to an effective SNR and compared with a fixed 13 dB decision level. Both stages use one configuration fixed before evaluation and shared across all turbulence regimes. This information flow uses only quantities available before either carrier-recovery branch produces decoded bits. The resulting study makes three contributions. First, it states reproducible signal, block, pilot, and total-energy definitions that distinguish the 100-symbol atmospheric channel block from the 256-sample receiver window. Second, it develops an interpretable two-stage rule that combines window-level power variation with a blind effective-SNR estimate and evaluates the resulting selector without regime-specific retuning. Third, it quantifies the complementary operating regions of the fixed DA and NDA estimators on both data-symbol-SNR and total-energy coordinates. In the predefined high-SNR work region, the data-symbol-SNR gaps are approximately 1.3, 1.2, and 1.9 dB for strong downlink, moderate uplink, and strong uplink

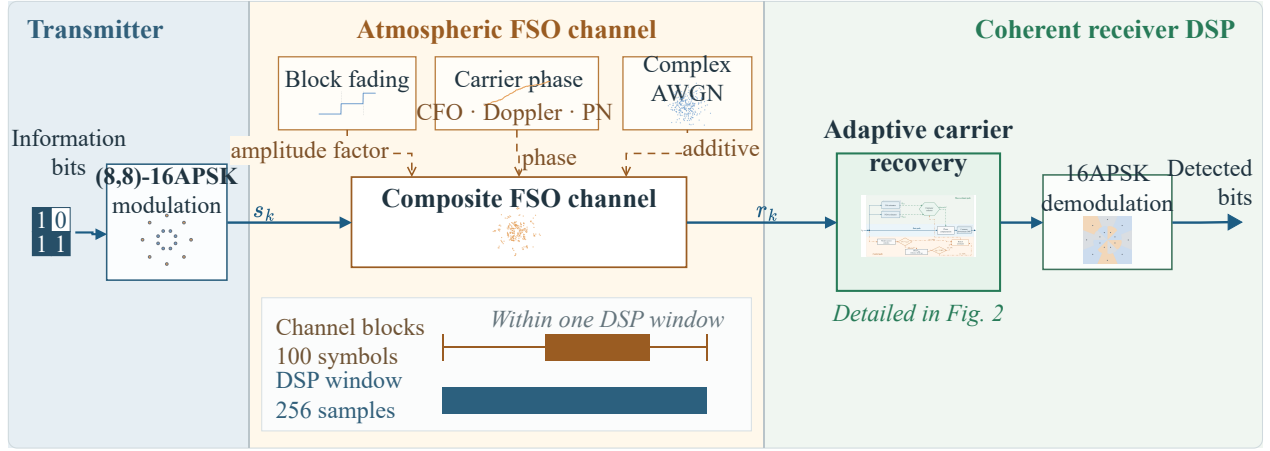


Fig. 1. System model of the coherent satellite-ground FSO link.

turbulence, respectively. Including the 1.249 dB DA pilot-energy term gives corresponding total-energy-equivalent NDA advantages of approximately 2.5, 2.4, and 3.1 dB.

The remainder of the paper is organized as follows. Section II defines the signal impairments, atmospheric and receiver block structure, pilot pattern, and SNR convention. Section III describes the DA and NDA estimators, the two-stage selection statistics, and the fixed decision parameters. Section IV presents the simulation setup, the complementary DA/NDA operating regions, and the selector evaluation. Section V concludes the paper.

II. SYSTEM MODEL

A. Signal and Impairment Model

As shown in Fig. 1, we consider a coherent free-space optical (FSO) link between a satellite and an optical ground station, including both downlink and uplink propagation regimes. The transmitted symbols use (8,8) 16-ary amplitude-phase shift keying (16APSK), with eight equally spaced phase points on each of two amplitude rings, following the modulation definition used by Du *et al.* [11]. The constellation is normalized so that its average symbol energy is $E_s = 1$.

Let $b(k) = \lfloor k/N_{\text{ch}} \rfloor$ identify the atmospheric channel block containing symbol k , where $N_{\text{ch}} = 100$ symbols. The discrete-time complex-baseband observation is

$$r_k = \sqrt{h_{b(k)}} s_k \exp(j\phi_k) + n_k, \quad (1)$$

where s_k is the transmitted symbol, $h_{b(k)}$ is the normalized irradiance, or intensity gain, and $n_k \sim \mathcal{CN}(0, N_0)$ is circular complex Gaussian noise. The square root is required because irradiance scales signal power, whereas the complex-baseband coefficient scales amplitude. Consequently, the instantaneous received signal-to-noise ratio (SNR), before receiver estimation error, is $h_{b(k)}\bar{\gamma}$, where $\bar{\gamma} = E_s/N_0$ is the average pre-fading SNR.

The irradiance follows a Gamma-Gamma model [5], [16] and is constant only within each 100-symbol channel block. It

is generated as the product of two unit-mean Gamma variates. The shape-parameter pairs $(\alpha, \beta) = (4.0, 3.0)$, $(2.5, 1.8)$, and $(1.5, 0.8)$ define the weak, moderate, and strong downlink simulation regimes; $(1.2, 0.9)$ and $(1.0, 0.7)$ define the moderate and strong uplink regimes. These parameter pairs define the numerical fading regimes evaluated below. Treating the atmospheric gain as constant over a short processing interval is consistent with reported FSO timing analyses that place scintillation dynamics on a substantially slower time scale than a microsecond-scale coherent frame [8], [17].

Within each evaluated 256-sample processing window, the carrier phase evolves continuously and contains a residual frequency offset f_{res} , a linear Doppler rate $\dot{f}$, and laser phase noise:

$$\phi_k = \phi_{\text{init}} + 2\pi f_{\text{res}} k T_s + \pi \dot{f} (k T_s)^2 + \sum_{\ell=0}^k \epsilon_{\ell}, \quad (2)$$

$$\epsilon_{\ell} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 2\pi \Delta\nu T_s),$$

where T_s is the symbol interval, $\Delta\nu = 10$ kHz is the combined laser linewidth, and ϕ_{init} is an arbitrary initial phase. The Wiener-increment variance follows the standard linewidth model used for coherent FSO carrier estimation [11]. Unlike the atmospheric coefficient, this phase is not assumed constant within a channel block.

For the turbulence simulations, each 256-sample processing window uses an independently initialized phase realization. Equation (2) therefore specifies continuous phase evolution within a processing window, while successive simulated windows provide independent realizations for Monte Carlo evaluation.

Equations (1) and (2) keep the impairment roles explicit. The irradiance term controls received signal power, the residual frequency and Doppler-rate terms produce deterministic phase evolution, the Wiener term produces stochastic phase diffusion, and n_k sets the additive-noise floor. In particular, neither the random irradiance nor the carrier phase is absorbed

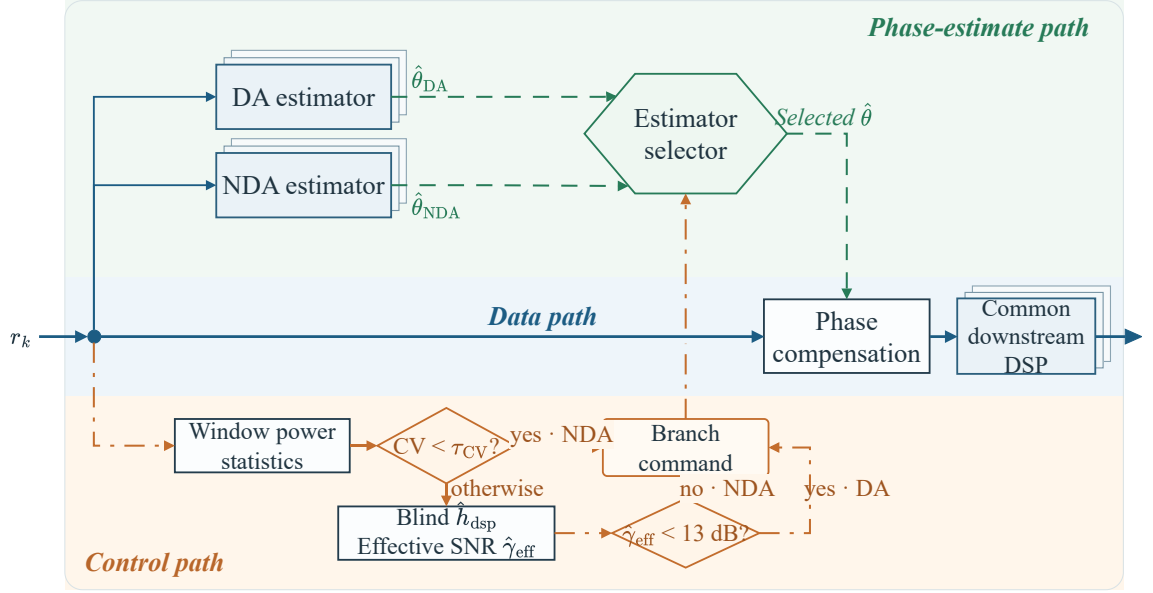


Fig. 2. Received-power-aware DA/NDA carrier phase recovery with two-stage estimator selection.

into $\bar{\gamma}$. This convention lets all receiver branches process the same channel realization while differing only in the observations and estimators available to them.

B. Block Structure, Pilot Pattern, and SNR Definition

Two distinct partitions are used. The atmospheric channel block has $N_{ch} = 100$ symbols and determines where $h_{b(k)}$ is held constant. Carrier recovery and estimator selection operate on nonoverlapping digital signal processing (DSP) windows of $N_{DSP} = 256$ samples. Because these lengths differ, a DSP window may contain samples from multiple atmospheric channel blocks; the receiver does not assume a single value of h over all 256 samples. Channel-gain estimation is therefore performed on the 100-symbol partition, while phase estimation and method selection use the 256-sample partition.

Within each DSP window, the data-aided layout assigns local indices $0, 4, \dots, 252$ to 64 known pilots and the remaining 192 positions to data. Channel-block boundaries are applied independently of this local layout, so pilots contributing to a channel-gain estimate are selected by their global symbol indices rather than by treating the DSP window as one fading interval. The same transmitted symbols, atmospheric gains, phase sequence, and noise realization are supplied to all compared receiver branches.

For the data-aided branch, one known pilot is placed every $L_p = 4$ transmitted symbols. Thus pilots occupy $\rho_p = 1/4$ of transmitted positions, while the expansion relative to the three data symbols is $1/3$. Comparisons use a total-transmit-energy coordinate: sending three data symbols plus one pilot requires $4/3$ of the energy used by three data-only symbols at the same

per-symbol energy. The coordinate offset is therefore

$$\begin{aligned} \rho_p &= \frac{1}{4}, \\ \Delta_p &= 10 \log_{10} \left(\frac{4}{3} \right) = 1.249 \text{ dB}, \\ \gamma_{tot,DA}^{dB} &= \gamma_{data}^{dB} + \Delta_p. \end{aligned} \quad (3)$$

The non-data-aided branch carries data in every symbol and receives no such coordinate offset. For both branches, bit error rate (BER) is evaluated over information-bearing data positions; pilot positions are excluded from the data-aided denominator. Performance is referenced to the pre-decoding BER threshold 3.8×10^{-3} , corresponding to 7% overhead hard-decision forward error correction (HD-FEC) [18], [19].

III. CARRIER PHASE RECOVERY AND ESTIMATOR SELECTION

The receiver uses data-aided (DA) and non-data-aided (NDA) carrier recovery as two complementary processing branches. Both branches operate on a digital-signal-processing window of $N_{DSP} = 256$ received samples, which is distinct from the 100-symbol atmospheric channel block defined in the system model. A branch is selected once per 256-sample window. The selection statistic is formed directly from the raw received samples, before branch-specific channel equalization and carrier recovery, so that the decision follows the same information flow for every turbulence regime. After selection, the DA path uses pilot-derived channel information, whereas the NDA path uses a received-power-based channel estimate. Figure 2 summarizes the parallel carrier-recovery branches and the two-stage control path described below.

A. DA and NDA Carrier Estimation

The DA branch uses one known pilot every four transmitted symbols. Let n_ℓ denote the index of the ℓ th pilot in the current processing window, p_ℓ its known complex value, and $t_\ell = n_\ell T_s$ its sampling time. Removing the known pilot modulation and fitting a linear phase model gives

$$\begin{aligned} \vartheta_\ell &= \text{unwrap} \left\{ \angle \left(\frac{r_{n_\ell}}{p_\ell} \right) \right\}, & \vartheta_\ell &\simeq \phi_0 + 2\pi \Delta f t_\ell, \\ \widehat{\Delta f}_{\text{DA}} &= \frac{\sum_\ell (t_\ell - \bar{t})(\vartheta_\ell - \bar{\vartheta})}{2\pi \sum_\ell (t_\ell - \bar{t})^2}, \\ \hat{\phi}_{0,\text{DA}} &= \bar{\vartheta} - 2\pi \widehat{\Delta f}_{\text{DA}} \bar{t}. \end{aligned} \quad (4)$$

Thus, the same set of regularly spaced pilots supplies both the frequency slope and the carrier-phase intercept. The compensated DA samples are obtained by removing $\hat{\phi}_{0,\text{DA}} + 2\pi \widehat{\Delta f}_{\text{DA}} k T_s$ from the received phase. The pilot density and its associated total-energy signal-to-noise ratio (SNR) convention follow the system-model definition; the present section uses that convention without introducing a second SNR coordinate.

With a spacing of four, a 256-sample window contains 64 pilot positions at indices $0, 4, \dots, 252$. The regression treats their unwrapped phases as one ordered sequence rather than as independent single-pilot estimates. Centering the pilot times in Eq. (4) gives the frequency slope from their covariance with phase, while the intercept refers the phase estimate to the beginning of the processing window. The receiver then constructs one phase ramp for all 256 samples. This makes the DA output a carrier-corrected complex sequence with the same length and sample timing as the input window, which allows the downstream demodulator to use the same interface for either selected branch.

The NDA branch removes the (8, 8)-16APSK modulation periodicity through the M_0 th-power operation, with $M_0 = 8$, complementing data-aided and blind phase search approaches that combine pilot symbols with modulation-removal estimators [20]. A frequency estimate is first obtained from the spectrum of the raised signal and removed from the received window. Carrier phase is then estimated from the argument of the averaged M_0 th-power samples. For a sample set $\mathcal{I}$, the basic estimate is

$$\hat{\phi}_{\mathcal{I}} = \frac{1}{M_0} \angle \left(\frac{1}{|\mathcal{I}|} \sum_{k \in \mathcal{I}} r_k^{M_0} \right), \quad M_0 = 8. \quad (5)$$

Two windowing modes implement this operation. Under Gamma–Gamma turbulence, $\mathcal{I}$ contains the complete 256-sample processing window and produces one window-level carrier-phase estimate. In the additive white Gaussian noise configuration, the window is divided into $K = 8$ contiguous segments of 32 samples. An M_0 th-power mean-angle estimate is computed at each segment center, the eight raised-domain phases are unwrapped across the window, and linear interpolation produces a sample-wise phase trajectory before division by M_0 . This segmented form follows the intra-window Wiener phase evolution while retaining the same

modulation-removal operation as the whole-window form. In both modes, the recovered phase is applied through complex derotation of the corresponding received samples. The DA and NDA branches therefore differ in their available reference information and phase construction, while sharing the same 256-sample decision interval.

The NDA frequency stage precedes these phase modes. Raising the samples to the eighth power removes the ring-wise angular periodicity and produces a spectral component whose frequency is eight times the carrier-frequency offset. After locating that component, the receiver divides the raised-domain estimate by M_0 , removes the corresponding phase ramp, and applies either the whole-window or segmented phase construction to the frequency-compensated samples. The order is consequently modulation removal, frequency estimation, frequency compensation, and carrier-phase estimation. The resulting derotation preserves the sample order and returns a 256-sample complex window for demodulation.

The eighth-power phase statistic is defined modulo $2\pi/8$. For the reported NDA bit-error curves, this eightfold phase ambiguity is resolved separately in each processing window by testing the eight constellation rotations against the transmitted bit labels and retaining the rotation with the lowest error count. This transmitted-bit-assisted step is an ambiguity-resolved evaluation protocol: it is not an online input to the NDA phase statistic or to the selector. The NDA results therefore characterize carrier tracking after phase-ambiguity resolution.

B. Two-Stage Selection Rule

The switching controller processes the raw received power $q_k = |r_k|^2$ over the current 256-sample window. Its first statistic is the coefficient of variation (CV), which describes the normalized spread of the received power:

$$\begin{aligned} \text{CV} &= \frac{\text{std}(q_k)}{\text{mean}(q_k)}, \\ \tau_{\text{CV}}(\bar{\gamma}_{\text{dB}}) &= 1.10 \left(0.74 + 0.12e^{-\bar{\gamma}_{\text{dB}}/5} \right). \end{aligned} \quad (6)$$

Here, $\bar{\gamma}_{\text{dB}}$ is the nominal data SNR supplied to the receiver. The reference curve $0.74 + 0.12e^{-\bar{\gamma}_{\text{dB}}/5}$ was fitted to an offline AWGN characterization of the received-power statistic, and the factor 1.10 provides the fixed decision margin. A window with $\text{CV} < \tau_{\text{CV}}$ is directed to the NDA branch. Otherwise, the controller proceeds to the received-power proxy and effective-SNR stage.

For the second stage, let $\bar{\gamma} = 10^{\bar{\gamma}_{\text{dB}}/10}$ denote the linear nominal data SNR. The controller forms a blind power-based channel proxy, maps it to an effective SNR, and applies the complete branch rule:

$$\begin{aligned} \hat{h}_{\text{dsp}} &= \max \left\{ \frac{1}{N_{\text{DSP}}} \sum_{k=0}^{N_{\text{DSP}}-1} |r_k|^2 - \frac{1}{2\bar{\gamma}}, 10^{-6} \right\}, \\ \hat{\gamma}_{\text{eff,dB}} &= \bar{\gamma}_{\text{dB}} + 10 \log_{10} \hat{h}_{\text{dsp}}, \\ \mathcal{A}(r) &= \begin{cases} \text{NDA}, & \text{CV} < \tau_{\text{CV}}, \\ \text{DA}, & \text{CV} \geq \tau_{\text{CV}} \text{ and } \hat{\gamma}_{\text{eff,dB}} < 13 \text{ dB}, \\ \text{NDA}, & \text{CV} \geq \tau_{\text{CV}} \text{ and } \hat{\gamma}_{\text{eff,dB}} \geq 13 \text{ dB}. \end{cases} \end{aligned} \quad (7)$$

Equation (7) makes the execution order explicit. The CV comparison is evaluated first. Only windows that pass this first boundary use $\hat{h}_{\text{dsp}}$ and $\hat{\gamma}_{\text{eff,dB}}$ for the 13-dB comparison. This arrangement separates windows whose raw power variation satisfies the offline-characterized first-stage condition from those for which the effective-SNR decision distinguishes the DA and NDA operating regions.

Operationally, the controller first accumulates q_k across the received window and forms its mean and standard deviation. The normalized spread is compared with τ_{CV} ; an NDA decision at this stage completes the route for that window. For the remaining windows, the already accumulated mean power is reused in $\hat{h}_{\text{dsp}}$, combined with the nominal data SNR, and transformed into $\hat{\gamma}_{\text{eff,dB}}$. The final comparison sends values below 13 dB to DA and values at or above 13 dB to NDA. This staged data flow implements the piecewise rule directly and keeps the first- and second-stage decision variables distinct.

C. Online Statistics and Fixed Parameters

The controller separates quantities evaluated for every received window from quantities fixed before operation. The online sample set consists of the 256 raw complex samples r_k . Their squared magnitudes provide the sample mean and standard deviation used by the CV stage, and the same mean power enters $\hat{h}_{\text{dsp}}$. The nominal data SNR $\bar{\gamma}_{\text{dB}}$ is an input to the controller: it selects the value of the CV boundary, determines the noise-subtraction term in the blind power proxy, and provides the reference level for $\hat{\gamma}_{\text{eff,dB}}$. No recovered data symbols or branch-output bit errors enter the selection rule.

The 13-dB decision level was selected from the 12–14-dB effective-SNR transition region observed in preliminary diagnostic sweeps. After this characterization, the CV coefficients 0.74, 0.12, and 5, the margin factor 1.10, the proxy floor 10^{-6} , and the 13-dB level were fixed and applied unchanged across all turbulence regimes and Monte Carlo runs, without per-regime retuning. The sample mean, sample standard deviation, blind power proxy, and effective SNR are nevertheless recomputed for each processing window. This division gives a reproducible receiver interface: raw received power and nominal data SNR are the decision inputs; the CV boundary and the 13-dB level define the stored decision configuration; and the selected branch supplies the carrier-corrected samples to the subsequent demodulator. The implementation consequently realizes the sequence CV screening, blind-power evaluation, effective-SNR construction, and fixed-level branch selection stated in Eqs. (6)–(7).

Neither the turbulence label nor the realized channel gain is supplied to the selector. The stored parameters come only from the offline AWGN characterization and preliminary diagnostics, while online adaptation enters through the current window statistics. Consequently, the weak, moderate, and strong labels organize the evaluation but do not alter the receiver configuration. This separation prevents per-regime parameter choices from embedding knowledge of the evaluated condition and keeps the comparison tied to one receiver configuration across all evaluated conditions.

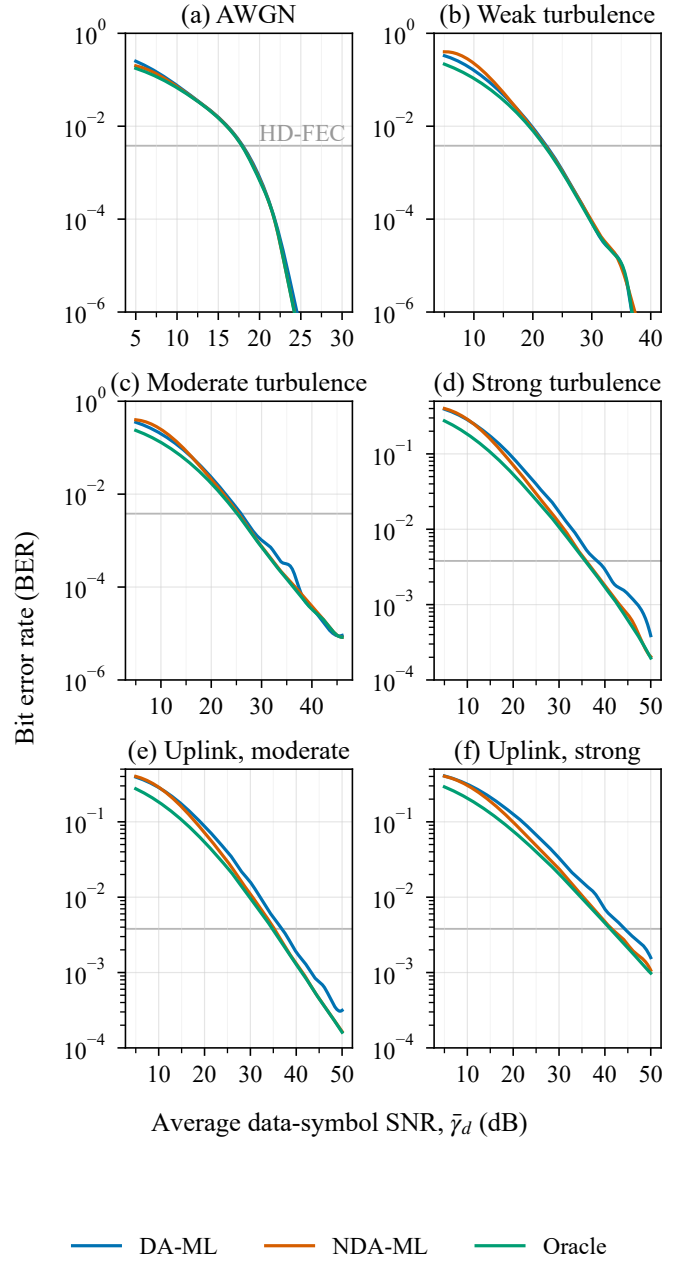


Fig. 3. Data BER of DA-ML, NDA-ML, and oracle phase recovery under AWGN and Gamma-Gamma turbulence.

IV. SIMULATION RESULTS

A. Simulation Setup and Performance Metrics

We evaluate the data-aided (DA) and non-data-aided (NDA) carrier-phase-recovery paths over six channel conditions: an additive white Gaussian noise (AWGN) reference, weak, moderate, and strong downlink Gamma-Gamma turbulence, and moderate and strong uplink turbulence. The benchmark sweep uses 30 independent seeds at every operating point. Each seed contains 400 processing windows of 256 received samples, or 102400 samples per seed and operating point. The

sampled average data-symbol signal-to-noise ratios (SNRs) span 5–20 dB for AWGN and 5–26 dB for all turbulent conditions. These common samples provide the basis for the fixed-estimator comparison. The selector-recovery count uses a separate fixed-configuration 30-seed data layer covering 29 prescribed points: eight AWGN SNRs and seven SNRs in each of the three downlink regimes.

Figure 3 also displays additional high-SNR samples so that the low-BER portions of the curves remain visible. Those extension samples use five independent seeds and extend the AWGN, weak, moderate, and strong/uplink curves to 30, 40, 46, and 50 dB, respectively. Together, the 30-seed benchmark and five-seed extension provide the complete displayed SNR range. Figure 4 uses the same merged curve set, while the selector count is evaluated on its separate fixed-configuration 30-seed grid.

The primary fixed-estimator metric is the bit error rate (BER), P_b . For NDA, all 256 16-APSK symbols carry data, giving 1024 data bits per processing window. With one pilot every four symbols, DA is evaluated over the remaining 192 data symbols, giving 768 data bits per processing window. This data-BER convention evaluates the decoded payload positions of each branch and is used to determine the lower-BER estimator in the 26-of-29 selector-recovery count. The 7% hard-decision forward-error-correction (HD-FEC) reference is $P_b = 3.8 \times 10^{-3}$ [18]; Fig. 3 uses it as a visual performance reference.

The oracle curves in Fig. 3 are performance references rather than candidate selector branches. The AWGN oracle removes the true carrier phase, while the turbulence oracle uses both the true irradiance and the true carrier phase. Neither oracle quantity enters the DA/NDA selector.

B. DA/NDA Operating Regions

Figure 3 shows the complementary operating regions of the two estimators. DA uses known pilot symbols to support carrier estimation, whereas NDA preserves all symbol positions for payload. In the predefined work region $\gamma_{\text{tot}} \geq 15$ dB, the average NDA-versus-DA gaps on the data-symbol-SNR coordinate are approximately 1.3 dB for strong downlink turbulence, 1.2 dB for moderate uplink turbulence, and 1.9 dB for strong uplink turbulence. Accounting for the DA pilot-energy term $\Delta_p = 10 \log_{10}(4/3) = 1.249$ dB maps these values to total-energy-equivalent advantages of approximately 2.5, 2.4, and 3.1 dB, respectively. These values compare the two fixed estimators and are distinct from the selector-output metric considered below.

The intersections of the DA and NDA data-BER curves provide a second view of this complementarity. Using linear interpolation in log-BER between simulated samples, the observed crossover SNRs are 18.0, 16.9, and 10.7 dB for weak, moderate, and strong downlink turbulence, respectively. Figure 4 marks these post-simulation curve observations directly. The receiver uses one fixed 13 dB effective-SNR decision level shared by all evaluated regimes.

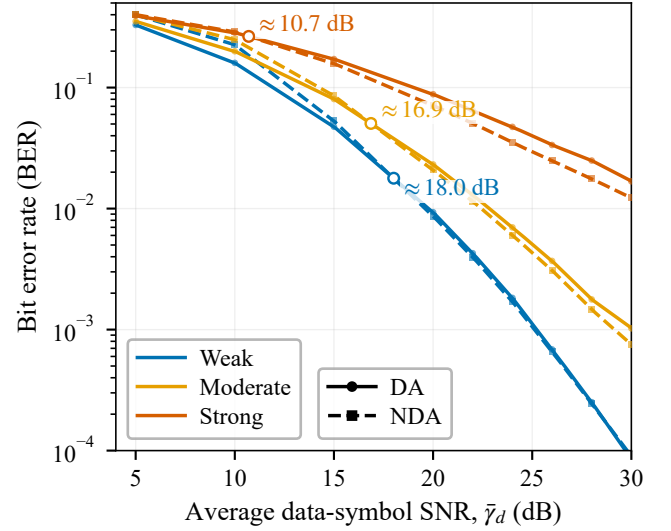


Fig. 4. DA-NDA data-BER crossovers under downlink turbulence.

C. Selector Performance

We quantify the selector relative to fixed NDA at the same average data-symbol SNR using a full-block-normalized error ratio. For branch $x \in \{\text{NDA}, \text{SW}\}$, let $R_{\text{FB},x} = N_{\text{err},x}/(1024N_{\text{win}})$, where $N_{\text{err},x}$ is the recorded bit-error count over N_{win} processing windows. The logarithmic error-ratio reduction is

$$G_{\text{FB}} = 10 \log_{10} \left(\frac{R_{\text{FB},\text{NDA}}}{R_{\text{FB},\text{SW}}} \right), \quad (8)$$

This normalization is distinct from the branch-specific data-BER denominators used for the selector-recovery count. A positive G_{FB} gives the logarithmic reduction of the full-block-normalized error ratio at a common data-symbol SNR. Figure 5 reports this metric over the three downlink turbulence regimes. Representative positive values are 2.3 dB at 10 dB in weak turbulence, 2.0 dB at 10 dB in moderate turbulence, and 1.3 dB at 5 dB in strong turbulence. These anchors illustrate the reduction obtained when the selector activates the branch favored by its fixed decision rule.

Across the selector grid, 29 operating points are evaluated: eight AWGN SNRs plus seven SNRs in each of the weak, moderate, and strong downlink regimes, i.e., $29 = 8 + 7 + 7 + 7$. Using the standard data-BER definition to identify the lower-BER fixed estimator, the selector recovers that estimator at 26 of the 29 points. This count and Fig. 5 describe complementary aspects of operation: the count measures branch recovery using the branch-specific payload denominators, whereas G_{FB} measures the full-block-normalized error-ratio reduction relative to fixed NDA at each common SNR. Together, they show that a single fixed 13 dB decision level shared across regimes converts the DA-NDA operating-region complementarity into repeatable estimator selection.

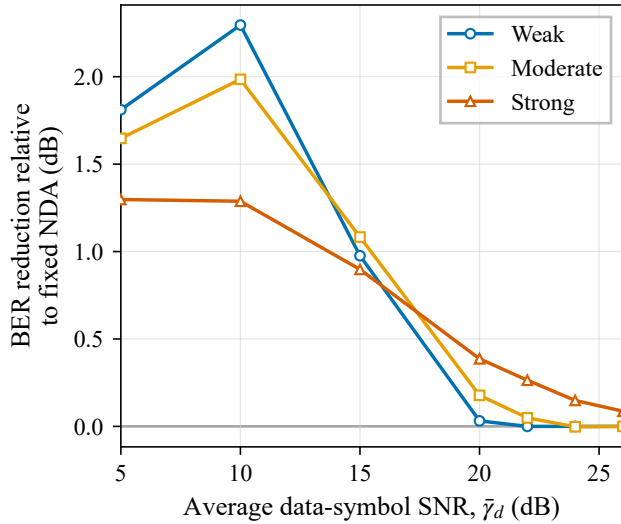


Fig. 5. Same-SNR full-block-normalized error-ratio reduction of received-power-aware switching relative to fixed NDA.

V. CONCLUSION

This paper presented a two-stage DA/NDA estimator selector for carrier phase recovery in turbulent satellite-ground FSO links. A coefficient-of-variation gate identifies windows with pronounced power variation, after which a received-power-based irradiance proxy and a fixed 13 dB effective-SNR level select between multi-pilot phase-time least squares and eighth-power NDA estimation. The design uses received-window statistics available before carrier recovery and applies one fixed configuration across the evaluated turbulence regimes. In the predefined high-SNR work region, fixed NDA provides a total-energy-equivalent advantage of up to 3.1 dB over fixed DA, comprising a 1.9 dB data-symbol-SNR gap and the approximately 1.25 dB DA pilot-energy term. Together with the observed DA-NDA crossovers, these results support a compact and interpretable approach to matching the carrier phase estimator to changing atmospheric conditions.

REFERENCES

- [1] R. Le Bidan, H. Méric, and J. Sommer, "Frame format and DSP receiver design for a 56-GBaud GEO DP-QPSK coherent optical feeder link," in *2023 IEEE International Conference on Space Optical Systems and Applications (ICSOS)*. Vancouver, BC, Canada: IEEE, 2023, pp. 77–84.
- [2] C. Valjus, R. Wolf, and J. Poliak, "Review and analysis of digital signal processing algorithms for coherent optical satellite links," *International Journal of Satellite Communications and Networking*, vol. 43, no. 3, pp. 229–250, 2025.
- [3] L. Paillier, R. L. Bidan, J.-M. Conan, G. Artaud, N. Vedrenne, and Y. Jaouen, "Space-ground coherent optical links: Ground receiver performance with adaptive optics and digital phase-locked loop," *Journal of Lightwave Technology*, vol. 38, no. 20, pp. 5716–5727, 2020.
- [4] F. P. Guiomar, M. A. Fernandes, J. L. Nascimento, V. Rodrigues, and P. P. Monteiro, "Coherent free-space optical communications: Opportunities and challenges," *Journal of Lightwave Technology*, vol. 40, no. 10, pp. 3173–3186, 2022.
- [5] M. A. Al-Habash, L. C. Andrews, and R. L. Phillips, "Mathematical model for the irradiance probability density function of a laser beam propagating through turbulent media," *Optical Engineering*, vol. 40, no. 8, pp. 1554–1562, 2001.
- [6] S. Pech, F. Destic, A. Dion, and A. Rissons, "Z-transform model of a coherent receiver for satellite-to-ground laser links under high doppler rates," in *2025 IEEE International Conference on Space Optical Systems and Applications (ICSOS)*. IEEE, 2025, pp. 1–10.
- [7] M. G. Taylor, "Phase estimation methods for optical coherent detection using digital signal processing," *Journal of Lightwave Technology*, vol. 27, no. 7, pp. 901–914, 2009.
- [8] A. Johst, L. Molle, N. Perlot, M. Rothe, M. Rohde, and M. Nölle, "Data-aided multi-format DSP for robust free-space coherent optical communication," in *2024 IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE)*. IEEE, 2024, pp. 82–87.
- [9] J. Panasiewicz, N. Arab, F. Destic, G. M. Pacheco, and A. Rissons, "All-digital optical phase-locked loop for satellite communications under turbulence effects," in *2022 International Topical Meeting on Microwave Photonics (MWP)*. IEEE, 2022, pp. 1–4.
- [10] A. J. Viterbi and A. M. Viterbi, "Nonlinear estimation of PSK-modulated carrier phase with application to burst digital transmission," *IEEE Transactions on Information Theory*, vol. 29, no. 4, pp. 543–551, 1983.
- [11] X. Du, K. Yu, Q. Wang, and P.-Y. Kam, "Non-data-aided ML estimation of timing offset and carrier phase for M-APSK modulated FSO systems," *IEEE Photonics Technology Letters*, vol. 37, no. 10, pp. 559–562, 2025.
- [12] W. Ma, S. Liu, X. Du, Q. Wang, and P.-Y. Kam, "Maximum a posteriori probability phase recovery for 256-QAM modulated signals in coherent optical communications," in *2025 30th OptoElectronics and Communications Conference and 2025 International Conference on Photonics in Switching and Computing (OECC/PSC)*. IEEE, 2025, pp. 1–4.
- [13] Q. Wang, Z. Quan, S. Bi, and P.-Y. Kam, "Joint ML/MAP estimation of the frequency and phase of a single sinusoid with wiener carrier phase noise," *IEEE Transactions on Signal Processing*, vol. 70, pp. 337–350, 2022.
- [14] X. Du, T. Song, Y. Li, M.-W. Wu, and P.-Y. Kam, "An optimum signal detection approach to the joint ML estimation of timing offset, carrier frequency and phase offset for coherent optical OFDM," *Journal of Lightwave Technology*, vol. 39, no. 6, pp. 1629–1644, 2021.
- [15] C. Wang, Z. Li, Z. Dong, R. Gao, J. Hao, J. Song, D. Guo, F. Wang, T. Mai, X. Pan, H. Chang, S. Zhou, and X. Xin, "Fully utilized pilot-aided digital signal processing with state pruning MLSD assisted adaptive symbol detection for symbol-rate sampling coherent optical interconnects," *Optics Express*, vol. 33, no. 14, pp. 29 026–29 043, 2025.
- [16] N. Liu, C. Ju, D. Wang, D. Wang, and P. Xie, "Multi-aperture coherent digital combining based on complex-valued MIMO $2N \times 2$ adaptive equalizer for FSO communication," *Journal of Lightwave Technology*, vol. 41, no. 18, pp. 5983–5990, 2023.
- [17] P. Conroy, J. Surof, J. Poliak, and R. Mata Calvo, "Demonstration of 40 GBaud intradyne transmission through worst-case atmospheric turbulence conditions for geostationary satellite uplink," *Applied Optics*, vol. 57, no. 18, pp. 5095–5101, 2018.
- [18] B. P. Smith, A. Farhood, A. Hunt, F. R. Kschischang, and J. Lodge, "Staircase codes: FEC for 100 Gb/s OTN," *Journal of Lightwave Technology*, vol. 30, no. 1, pp. 110–117, 2012.
- [19] Y. Horst, B. I. Bitachon, L. Kulmer, J. Brun, T. Blatter, J.-M. Conan, A. Montmerle-Bonnefois, J. Montri, B. Sorrente, C. B. Lim, N. Vedrenne, D. Matter, L. Pommarel, B. Baeuerle, and J. Leuthold, "Tbit/s line-rate satellite feeder links enabled by coherent modulation and full-adaptive optics," *Light: Science & Applications*, vol. 12, no. 1, p. 153, 2023.
- [20] C. S. Martins, F. P. Guiomar, and A. N. Pinto, "Hardware optimization of dual-stage carrier-phase recovery for coherent optical receivers," *OSA Continuum*, vol. 4, no. 12, pp. 3157–3175, 2021.